Computer Architecture and Operating Systems
COSC 22073/BECS 22233

Lesson 05: Memory Organization

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What is computer memory organization?

- Computer memory organization refers to how a computer's memory is structured and managed to store and retrieve data. Memory in a computer system can be organized into several different levels, each with its own characteristics and purpose

Computer Memory - Basic Concepts

- Ideally, the memory would fast, large and inexpensive.
- Unfortunately, it is impossible to meet all three of these requirements simultaneously.
- Increased speed and size are achieved at increased cost.
- To solve this problem, much work has gone into developing clever structures that improve the apparent speed and size of the memory, yet keep the cost reasonable.

Computer Memory - Basic Concepts

- The maximum size of the memory can be determined by the addressing scheme.

  Ex: a 16-bit computer that generates 16-bit addresses is capable of addressing up to 216 = 64K memory locations.

- The memory is usually designed to store and retrieve data in word-length quantities.
- The number of bits actually stored or retrieved in one memory access is the most common definition of the word length of a computer.

Computer Memory - Basic Concepts

- We can view the memory unit as a black box.
- Data transfer between the memory and the processor takes place through the use of two processor registers, called MAR and MDR.

Memory Address Register (MAR):

1. The Memory Address Register (MAR) is a processor register that holds the address of the memory location being read from or written to. It specifies the memory location that the CPU wants to access.

Memory Data Register (MDR):

1. The Memory Data Register (MDR) is another processor register that holds the data being transferred between the CPU and memory. It stores the actual data that is read from memory or the data that the CPU intends to write into memory.

Computer Memory - Basic Concepts

- A useful measure of the speed of memory units is the time that elapses between the initiation of an operation and the completion of that operation (Memory access time).
- Memory cycle time – Minimum time delay required between the initiation of two successive memory operations.
- The cycle time is usually slightly longer than the access time, depending on the implementation details of the memory unit.
- A memory unit is called **Random Access Memory (RAM)** if any location can be accessed for a Read or Write operation in some fixed amount of time that is independent of the location's address.

Computer Memory - Basic Concepts

- The basic technology for implementing the memory, uses semiconductor integrated circuits.
- To reduce the memory access time it can be used a cache memory.
- Cache Memory – A small, fast memory that is inserted between the larger, slower main memory and the processor.
- It holds the currently active segments of a program and their data.

Computer Memory - Basic Concepts

- Virtual Memory – So far, we have assumed that the addresses generated by the processor directly specify physical locations in the memory.
- But there may be some cases which store data in physical memory locations that have addresses different from those specified by the program.
- The memory control circuitry translates the address specified by the program into an address that can be used to access the physical memory.

Computer Memory - Basic Concepts

- In such a case, an address generated by the processor is referred to as a virtual or logical address.
- The virtual address space is mapped onto the physical memory where data are actually stored.
- The mapping function is implemented by a special memory control circuit, often called the **memory management unit.**
- This mapping function can be changed during program execution according to system requirements.

Computer Memory - Basic Concepts

- Virtual memory is used to increase the apparent size of the physical memory.
- Data are addressed in a virtual address space that can be as large as the addressing capability of the processor.
- But at any given time, only the active portion of this space is mapped onto locations in the physical memory.
- The remaining virtual addresses are mapped onto the bulk storage devices used, which are usually magnetic disks.

Computer Memory - Basic Concepts

- As the active portion of the virtual address space changes during program execution, the memory management unit changes the mapping function and transfers data between the disk and the memory.
- Memories that consist of circuits capable of retaining their state as long as power is applied.
- Static RAMs are fast but Very expensive because their cells require several transistors

Computer Memory - Dynamic RAMs (Asynchronous)

- Less expensive RAMs can be implemented if simpler cells are used.
- However, such cells do not retain their state indefinitely.
- Hence they are called DRAMs.
- Information is stored in a dynamic memory cell in the form of a charge on a capacitor.
- This charge can be maintained for only tens of milliseconds.
- Since the cell is required to store information for a much longer time, its contents must be periodically refreshed by restoring the capacitor charge to its full value.
- DRAMs operation is directly synchronised with a clock signal.

Computer Memory Considerations

- The choice of a RAM chip for a given application depends on several factors.
  - Cost
  - Speed
  - Power dissipation
  - Size of the chip
- Static RAMs are used only when very fast operation is the primary requirement.
- Their cost and size are adversely affected by the complexity of the circuit that realises the basic cell.
- They are mostly used in cache memories.

Computer Memory Considerations

- Dynamic RAMs are the predominant choice for implementing computer main memories.
- The high densities achievable in these chips make large memories economically feasible.

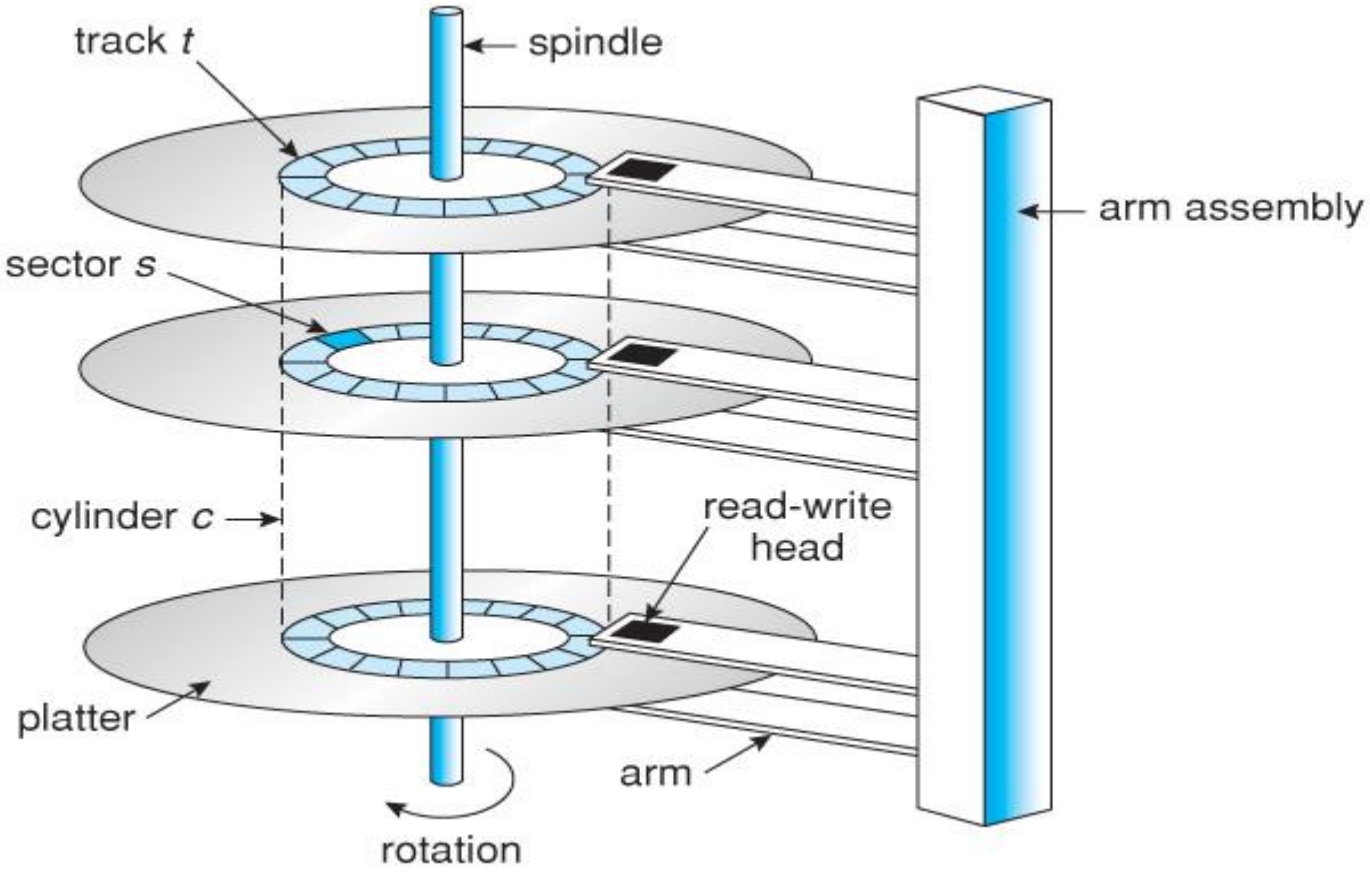
Computer Memory - Secondary Storage

- Semiconductor memories cannot be used to provide all of the storage capability needed in computers.
- Their main limitation is the cost per bit of stored information.
- Large storage requirements of most computer systems are economically realised the secondary storage devices.
  - magnetic disks
  - optical disks
  - magnetic tapes

Secondary Storage - Magnetic Hard Disks

- Storage medium in a magnetic disk system consists of one or more disks mounted on a common spindle.
- A thin magnetic film is deposited on each disk, usually on both sides.
- The disks are placed in a rotary drive so that the magnetised surfaces move in close proximity to read/write heads.
- The disks rotate at a uniform speed.
- Each head consists of a magnetic yoke and a magnetising coil
- Digital information can be stored on the magnetic film by applying current pulses of suitable polarity to the magnetising coil.
- The same head can be used for reading the stored information.

Secondary Storage - Magnetic Hard Disks



Thank You.